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ENVELOPE 1 · TECHNICAL PROPOSAL

Supply, Installation, Commissioning and Maintenance of High-Energy Cargo X-Ray Inspection Systems

Port of Al-Noor · Tender NGPA/T/2026/014

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Submitted to	Northern Gulf Ports Authority, Tender Committee Secretariat
Bidder	**Harborline Systems B.V.** (Netherlands)
Our reference	HLS/NGPA/2026/014-T
Date	10 December 2026
Contact	Eva Lindqvist, Bid Manager, Middle East · bids@harborline.example · +31 10 000 0000

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Form F-1 — Bid submission letter (technical)

HLS/NGPA/2026/014-T · 10 December 2026

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for tender NGPA/T/2026/014, including **Addendum No. 1 dated 28 October 2026**, receipt of which we hereby acknowledge, we offer to supply, install, commission and maintain the High-Energy Cargo X-Ray Inspection Systems for the Port of Al-Noor in conformity with the tender documents.

Harborline Systems B.V. is the Original Equipment Manufacturer of the HX-series inspection systems offered. Our technical proposal comprises this document and its annexes; it contains no pricing information, in accordance with RFP §2.4.2. Our financial proposal is submitted separately in Envelope 2.

We confirm that our bid is valid for 120 days from the bid closing date of 14 December 2026, that we have signed the Authority's integrity pledge, and that we accept the Draft Contract (Annex C) including its Special Conditions without reservation.

For and on behalf of Harborline Systems B.V.
Pieter van Doorn, Managing Director

Date: 10 December 2026
Company seal

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1. Company profile and experience

Harborline Systems designs and manufactures high-energy X-ray inspection systems for ports, land borders and free zones. Founded in 1998, the company employs 1,150 people, of whom 310 work in research and development, and operates a factory and radiation test bunker in Rotterdam and a regional service hub serving the Gulf. Since 2012 Harborline has delivered 96 high-energy cargo scanners in 31 countries.

Harborline is the OEM of all inspection systems offered (Form F-3 not applicable). The detector arrays, image chain and HX Sentinel software are developed in-house; the linear accelerators are produced by our long-term supplier under a supply agreement that guarantees parts for 15 years.

Form F-4 — Relevant references (last seven years)

Site	Scope	Commissioned	Climate	End-user contact
Port of Mirsal, Container Terminal 2	3 × HX-6 drive-through, central analysis centre, 18 workstations	2021	Mean summer max 43 °C, RH 76 %	Head of Customs Inspection, Mirsal Customs
Kharif Bay Container Terminal	2 × HX-6 drive-through + 1 × HX-6R relocatable	2022	Mean summer max 41 °C, RH 72 %	Port Security Director, Kharif Bay Port Co.
Al-Wadi land border crossing	2 × HX-6 drive-through, integration with national risk engine	2023	Mean summer max 45 °C, RH 30 %	Director of Border Technology, Customs Authority
Port of Sela, Ro-Ro terminal	1 × HX-6 drive-through on a constrained plot (66 m × 27 m)	2024	Mean summer max 34 °C, RH 70 %	Terminal Manager, Sela Ro-Ro
Duqah Free Zone	2 × HX-6, 5-year comprehensive maintenance (ongoing)	2025	Mean summer max 44 °C, RH 74 %	Head of Operations, Duqah Free Zone Authority

Three of these installations are in seaports with a mean summer maximum above 40 °C and humidity above 70 % (Mirsal, Kharif Bay, Duqah). Contact details for all references are provided in the separately sealed reference envelope.

2. Solution overview

We offer three **HX-6 DualView** drive-through systems (T1 Gate A, T1 Gate B and T2 main gate) and one **HX-6R** relocatable system, all with a 6 MeV dual-energy linear accelerator, together with the **HX Sentinel** image management and analysis platform hosted at the Customs Data Center (CDC).

Site	System	Source	Views	Penetration (steel)	Throughput (driver in cab)
T1 Gate A	HX-6 DualView	6 MeV dual-energy linac	Top + side	320 mm	200 trucks/h

T1 Gate B	HX-6 DualView	6 MeV dual-energy linac	Top + side	320 mm	200 trucks/h
T2 main gate	HX-6 DualView (compact shielding)	6 MeV dual-energy linac	Top + side	320 mm	180 trucks/h
Relocatable	HX-6R	6 MeV dual-energy linac	Top	300 mm	150 trucks/h

Drivers remain in their cabs. The HX-6 uses a cab-gating sensor array that starts the high-energy beam only after the cab has passed the beam plane; the cab itself can be imaged by an optional separate low-dose 450 kV source (see §4).

3. Imaging and throughput performance

Parameter	T1 Gate A / B	T2	Relocatable	Verification
Steel penetration (ANSI N42.46-2008)	320 mm	320 mm	300 mm	FAT + SAT
Wire detection behind 100 mm steel	0.8 mm Cu	0.8 mm Cu	1.0 mm Cu	FAT
Contrast sensitivity	0.8 %	0.8 %	1.0 %	FAT
Material discrimination	5 classes (organic, light inorganic, inorganic, mixed, heavy metal)	5 classes	4 classes	FAT
Scan speed	5–15 km/h	5–15 km/h	5–12 km/h	SAT
Tunnel (H × W)	4.8 m × 3.2 m	4.8 m × 3.2 m	4.7 m × 3.1 m	Drawing

Throughput values are measured as 4-hour sustained averages with mixed 20' and 40' traffic at 8 km/h, matching the SAT method of Annex B. The T2 value is lower than at T1 because the shorter T2 approach limits queue spacing; it remains above the 150 trucks/h requirement. All lanes accept 45' high-cube containers on standard chassis.

The HX Sentinel software provides edge enhancement, density mapping, material colouring, and a library of 1,400 reference images for analyst training.

4. Site layouts, power and radiation safety

Dimensioned layout drawings are included in the drawings folio (drawings HLS-014-L01 to L05). The controlled area, including shielding walls and exclusion zones, fits every plot of Annex B, sheet "Site constraints":

Site	Plot (Annex B)	Controlled area offered	Shielding concept	Max. demand incl. HVAC	Available power
T1 Gate A	95 × 36 m	86 × 30 m	Modular steel-concrete walls 1.6 m	212 kVA	400 kVA
T1 Gate B	95 × 36 m	86 × 30 m	Modular steel-concrete walls 1.6 m	212 kVA	400 kVA

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T2 main gate	70 × 28 m	68 × 27 m	High-density concrete walls (3.4 t/m³), roof beam stop	238 kVA	250 kVA
Relocatable, T1 pad	80 × 32 m	74 × 30 m	Transportable shielding blocks	146 kVA	160 kVA
Relocatable, T2 pad	60 × 30 m	58 × 28 m	Transportable blocks + beam-stop container	146 kVA	100 kVA + 200 kVA generator (Harborline supply)

At the T2 relocatable pad the available 100 kVA is insufficient; a 200 kVA containerised diesel generator is included in our scope, as permitted by Annex B. No grid reinforcement is needed at T2 main gate, where our demand of 238 kVA stays within the 250 kVA available.

Radiation safety. The high-energy beam is never directed at an occupied cab (dual redundant cab-gating sensors, safety PLC to SIL 2). The optional cab scan uses a separate 450 kV source delivering **0.12 µSv per pass**. Annual dose at the controlled-area boundary is calculated at 0.6 mSv at 200 trucks/h (shielding calculation report HLS-014-R02). Emergency stops are placed so that one is reachable within 8 m from any point in the controlled area. Beam-on warning lights, audible alarms and bilingual (Arabic/English) signs are provided. Personal dosimeters are supplied for all operators during the first 12 months.

5. Radiation Licensing Support Plan (Addendum No. 1, A4)

Step	Activity	Responsible	Timing (month from effectiveness)
1	Pre-application meeting with the national radiation regulator	Harborline + Employer	M1
2	Shielding calculations, radiation protection programme, local rules	Harborline (Radiation Protection Adviser)	M1–M2
3	Licence application per system, signed by the Employer as licensee	Harborline prepares, Employer submits	M2 (T1 Gate A), M3 (others)
4	Import permit for the radiation sources	Harborline	M3
5	Commissioning surveys and regulator inspection	Harborline + regulator	Before each SAT
6	Licence issued; operation authorised	Regulator	Before trial operation of each system

No system is operated before it is licensed. A qualified Radiation Protection Adviser remains assigned until the last licence is issued.

6. Environmental design and power continuity

All outdoor equipment is type-tested for continuous operation from **−5 °C to +55 °C** at up to 95 % relative humidity in saline atmosphere (test certificate TC-HX-2023-07). Outdoor electronics enclosures are **IP55** with air-to-air heat exchangers; structural steel uses a C5-M coating system (zinc-rich epoxy primer, epoxy intermediate, polyurethane topcoat, 320 µm DFT) for a durability of more than 15 years. Structures are designed for a wind survival speed of **130 km/h**; scanning stops automatically above 60 km/h.

Supply: 400 V ±10 %, 3-phase, 50 Hz. UPS autonomy is **60 minutes** for control systems and servers (Addendum No. 1, A3) and **30 minutes** for workstations, with orderly shutdown that preserves images in progress. Noise at 1 m from operator cabins is below 65 dB(A).

7. Software, Central Image Analysis Centre and data

Hosting. HX Sentinel runs entirely on servers installed at the CDC and at each terminal. Licensing uses an offline licence file activated once by the Employer's staff (Addendum No. 1, Q3); no connection outside the Employer's network is required for operation, licence renewal, monitoring or updates. Updates are delivered as signed packages and installed by the Employer's administrators or by our engineers on site. No telemetry leaves the Employer's network.

Storage sizing (RFP §4.6.3, Annex B "Volumes & storage"):

Item	Value
Scans per year (all sites, 2028 volumes)	1,170,000
Average data per scan (both views + metadata)	45 MB
Data per year	52.65 TB
5-year online retention	263.25 TB
+ 20 % headroom	315.9 TB
Usable capacity offered (after RAID 6)	**340 TB**, expandable to 680 TB in the same racks

Images are stored in DICOS (NEMA IIC 1) natively and exported in DICOS, TIFF and JPEG. Workload-based routing sends each scan to the first free analyst at the CIAC or the terminals; the measured 90th-percentile scan-to-decision time at Port of Mirsal is **2.4 minutes**.

AI-assisted detection (RFP §4.7): HX Sentinel Detect highlights anomalies for undeclared goods, weapons, cigarettes, narcotics concealment patterns and currency. All models run on GPU servers at the CDC. The Employer's images are never used to train models outside the CDC; models fine-tuned at the CDC are the Employer's property (GC 12.2).

Backup. Daily encrypted backups to a second on-premise site designated by the Employer; RPO 24 hours, RTO 6 hours for the image platform.

8. Integration and OCR

CRMS: bidirectional REST/JSON over HTTPS (scan requests, results, thumbnails with image links, analyst annotations), with an OpenAPI specification delivered at design review. **TOS:** UN/EDIFACT COPARN and CODECO (D.95B) through the HX Integration Hub, or REST if agreed with the terminal operator. All middleware is included. Integration tests run against the Employer's test environments before SAT.

OCR: container numbers per ISO 6346 with check-digit validation (**99.1 %** correct reads at Port of Mirsal over 12 months) and licence plates in Arabic and Latin characters (**98.4 %**), with infrared illumination for night operation and wipers for dust.

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9. Cybersecurity

HX Sentinel complies with RFP §4.8 and with the Employer's IT Security & Data Residency Standard:

- Authentication through the Employer's Active Directory over LDAPS; role-based access control mapped to job functions; multi-factor authentication for all administrative and remote access (RADIUS integration with the Employer's MFA service).
- Tamper-evident (hash-chained) audit logs of all user and administrator actions, retained for 5 years and forwarded in real time to the Employer's SIEM in syslog/CEF format.
- Encryption at rest with AES-256 and in transit with TLS 1.3 (TLS 1.2 minimum), using certificates from the Employer's PKI.
- Software Bill of Materials (CycloneDX) for every release; quarterly security patches; critical vulnerabilities remediated within 14 days and high within 30 days, during warranty and CMC.
- Independent penetration test before go-live by an accredited third party; all critical and high findings closed before acceptance.
- Vendor remote access only through the Employer's VPN, approved per session, with full session recording. No permanent or vendor-initiated connections.
- Engineers with administrative access undergo the Directorate's security vetting and sign the confidentiality undertaking.

10. Workstations and user interface

We supply **20 workstations** (Addendum No. 1, A2): 10 at the CIAC, 5 at T1, 3 at T2 and 2 spare/mobile. Each has **two 27-inch colour-calibrated monitors at 2560 × 1440**, 64 GB RAM and an NVIDIA RTX-class GPU with 16 GB memory. The operator and analyst interfaces are fully available in Arabic and English, switchable per user, with dates shown in both Gregorian and Hijri calendars.

11. Implementation plan and acceptance

Milestone	Month
Design review and site surveys (incl. geotechnical survey at T2)	M1–M2
Civil works T1 Gate A	M2–M4
FAT of all systems in Rotterdam (4 Employer witnesses, travel at our cost)	M3–M4
T1 Gate A installed, SAT and start of trial operation	M5
T1 Gate B, T2 and relocatable SAT	M6–M7
CIAC platform, integration tests, penetration test	M5–M7
PAC of all systems after 30-day trial operation (≥ 95 % availability)	M9

Civil works are carried out by our local partner Al-Bina Contracting under Harborline's management. Method statements and risk assessments are submitted to the HSE Department before works start, and a full-time HSE officer is on site.

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12. Training and documentation

Audience	Participants	Duration	Location	Language
Scanner operators	40	5 days	On site	Arabic
Image analysts	12	10 days + 20 days on-the-job coaching	CIAC	Arabic
Maintenance engineers	6	15 days	Harborline factory, Rotterdam	English with simultaneous Arabic interpretation
System administrators	2	5 days	CDC	English

A train-the-trainer programme certifies four Employer instructors. Operation and maintenance manuals are delivered in English and Arabic, with as-built drawings and radiation survey reports.

13. Warranty, support and maintenance

Warranty: 24 months from the PAC of each system, covering all parts, labour and travel, including the linear accelerator and detector arrays; 36 months from PAC for workstations, servers, storage and network equipment (SC 9.2).

Service: Harborline Al-Noor Service Centre opens in the State three months before the first SAT with **six OEM-certified engineers** and an in-country stock of critical spare parts. Remote response within 2 hours and on-site response at T1/T2 within **12 hours**, 24/7.

Availability: guaranteed ≥ 98 % per scanner per calendar month (penalties per Annex C, Appendix 3). **CMC:** five years after warranty, including preventive and corrective maintenance, all spare parts, one linear accelerator replacement per system if needed (SC 18.4) and software updates. Spare parts are guaranteed for **15 years** after PAC, with 24 months' notice of any end-of-life.

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Annex 1. Form F-2 — Technical compliance matrix

One row per Annex B requirement, as amended by Addendum No. 1. "Reference" points to the section of this proposal.

Ref	Requirement	Status	Offered	Reference
TB-01	Steel penetration (ANSI N42.46): T1 A/B ≥ 310 mm, T2 ≥ 300 mm, relocatable ≥ 280 mm	Comply	320 mm (T1, T2); 300 mm (relocatable)	§3
TB-02	Wire detection behind 100 mm steel ≤ 1.0 mm Cu	Comply	0.8 mm Cu (fixed); 1.0 mm Cu (relocatable)	§3
TB-03	Dual-energy X-ray source	Comply	6 MeV dual-energy linac	§2
TB-04	Contrast sensitivity ≤ 1.0 %	Comply	0.8 % (fixed); 1.0 % (relocatable)	§3
TB-05	Material discrimination ≥ 4 classes	Comply	5 classes (fixed); 4 (relocatable)	§3
TB-06	Dual-view (top + side) imaging (optional for relocatable)	Comply	Top + side at T1 and T2; top only on relocatable	§2
TB-07	Throughput, driver in cab: T1 Gate A ≥ 180/h; T1 Gate B, T2 ≥ 150/h; relocatable ≥ 120/h (A5)	Comply	200/h T1 A/B; 180/h T2; 150/h relocatable	§3
TB-08	Vehicle speed during scan 5–15 km/h	Comply	5–15 km/h (relocatable 5–12)	§3
TB-09	Inspection tunnel ≥ 4.6 m × 3.0 m (H × W)	Comply	4.8 × 3.2 m (relocatable 4.7 × 3.1)	§3
TB-10	High-energy beam never on occupied driver cab	Comply	Redundant cab gating, SIL 2	§4
TB-11	Driver dose per pass ≤ 0.25 µSv (if cab scan offered)	Comply	0.12 µSv per pass (optional cab scan)	§4
TB-12	Annual dose at controlled-area boundary ≤ 1 mSv	Comply	0.6 mSv/yr calculated	§4
TB-13	Controlled area + exclusion zones fit the site plots	Comply	All plots; T2 68 × 27 m	§4
TB-14	Emergency stops, interlocks, beam-on lights, audible alarms	Comply	E-stop within 8 m; lights, alarms, signs	§4
TB-15	Continuous operating ambient 0 … +50 °C	Comply	−5 … +55 °C	§6
TB-16	Relative humidity ≤ 95 % non-condensing	Comply	≤ 95 %	§6
TB-17	Outdoor electronics enclosures ≥ IP54	Comply	IP55	§6
TB-18	Steel coating C5-M (ISO 12944)	Comply	C5-M, 320 µm	§6
TB-19	Wind survival 120 km/h	Comply	130 km/h	§6
TB-20	Supply 400 V ±10 %, 3-phase, 50 Hz	Comply	400 V ±10 %, 3-ph, 50 Hz	§6
TB-21	Maximum demand per system ≤ site capacity	Comply	212 / 212 / 238 / 146 kVA; generator at T2 pad	§4

TB-22	UPS autonomy, control systems and servers ≥ 60 min (A3)	**Comply**	60 min	§6
TB-23	UPS autonomy, workstations ≥ 30 min	**Comply**	30 min	§6
TB-24	On-premise hosting of all components incl. licensing, telemetry, updates	**Comply**	All on-premise; offline licence file; no telemetry	§7
TB-25	Data residency: in-country only	**Comply**	In-country only	§7
TB-26	Online retention of images and metadata: 5 years	**Comply**	5 years online	§7
TB-27	Usable storage at CDC per RFP §4.6.3 (state usable TB)	**Comply**	340 TB usable (required ≈ 316 TB)	§7
TB-28	DICOS (NEMA IIC 1) export	**Comply**	Native DICOS	§7
TB-29	CRMS bidirectional REST/JSON integration	**Comply**	REST/JSON, OpenAPI	§8
TB-30	TOS integration: EDIFACT COPARN/CODECO or REST	**Comply**	EDIFACT COPARN/CODECO D.95B	§8
TB-31	OCR container numbers (ISO 6346) ≥ 98 %	**Comply**	99.1 % (reference site)	§8
TB-32	OCR licence plates, Arabic + Latin ≥ 98 %	**Comply**	98.4 % Arabic + Latin	§8
TB-33	Scan-to-analyst-decision time, 90th percentile ≤ 3 min	**Comply**	2.4 min (reference site)	§7
TB-34	AD via LDAPS, RBAC, MFA for administrators	**Comply**	LDAPS, RBAC, RADIUS MFA	§9
TB-35	Tamper-evident audit log retention 5 years	**Comply**	5 years, hash-chained	§9
TB-36	Encryption AES-256 at rest / TLS ≥ 1.2 in transit	**Comply**	AES-256 / TLS 1.3	§9
TB-37	Vendor remote access via Employer VPN, per session, recorded	**Comply**	Employer VPN, per session, recorded	§9
TB-38	Operator/analyst UI in Arabic and English	**Comply**	Arabic + English	§10
TB-39	Hijri + Gregorian date display	**Comply**	Gregorian + Hijri	§10
TB-40	Monitors: 2 × 27", ≥ 2560 × 1440 per workstation	**Comply**	2 × 27", 2560 × 1440, calibrated	§10
TB-41	Workstations: 20 in total (A2)	**Comply**	20 workstations	§10
TB-42	Relocation T1 ↔ T2 incl. re-commissioning ≤ 72 h	**Comply**	48 hours	§2
TB-43	Availability per scanner per month ≥ 98 %	**Comply**	≥ 98 %	§13
TB-44	On-site response 24/7 ≤ 24 h	**Comply**	12 hours	§13
TB-45	Spare parts availability ≥ 10 years after PAC	**Comply**	15 years	§13

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Annex 2. Form F-6 — Statement of deviations (technical)

The Bidder declares that this proposal contains **no deviations** from the tender documents, including Addendum No. 1.

For and on behalf of Harborline Systems B.V.
Pieter van Doorn, Managing Director

Date: 10 December 2026
Company seal

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